

Ex.no:5

Implementation of Queue

Aim :

Design, Develop and Implement a menu driven Program in C for the following operations on queue of Integers

- a. Enqueue an Element on to queue
- b. Dequeue an Element from queue
- c. Demonstrate Overflow and Underflow situations on queue
- d. Display the status of queue
- e. Exit

Algorithm

Start

Initialize queue with front = -1, rear = -1, MAX size.

Menu Operations:

a) Enqueue (Insert):

- o If rear == MAX-1 → Overflow.
- o Else if front == -1 → set front = 0.
- o Increment rear → insert element at queue[rear].

b) Dequeue (Delete):

- o If front == -1 OR front > rear → Underflow.
- o Else remove element at queue[front], increment front.

c) Demonstrate Overflow / Underflow:

- o Handled during Enqueue and Dequeue checks.

d) Display Queue:

- o If front == -1 → Empty.
- o Else print elements from front to rear.

e) Exit:

- o Stop program.

Repeat menu until Exit.

Stop

Program

```
#include <stdio.h>
#include <stdlib.h>
#define MAX 5 // size of queue

int queue[MAX];
int front = -1, rear = -1;

// Function to Enqueue
void enqueue(int x) {
    if (rear == MAX - 1) {
        printf("\nQueue Overflow! Cannot insert %d\n", x);
        return;
    }
    if (front == -1) front = 0; // first element
    queue[++rear] = x;
    printf("\n%d inserted into queue.\n", x);
}
```

```

// Function to Dequeue

// Function to Display Queue

// Main Menu
int main() {
    int choice, item;

    do {
        printf("\n--- Queue Menu ---\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display Queue\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter element to insert: ");
                scanf("%d", &item);
                enqueue(item);
                break;

            case 2:
                dequeue();
                break;

            case 3:
                display();
                break;

            case 4:
                printf("\nExiting program.\n");
                break;

            default:
                printf("\nInvalid choice! Try again.\n");
        }
    } while (choice != 4);

    return 0;
}

```

Output

```

--- Queue Menu ---
1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Enter your choice: 1
Enter element to insert: 10
10 inserted into queue.

--- Queue Menu ---
Enter your choice: 1
Enter element to insert: 20

```

20 inserted into queue.

--- Queue Menu ---

Enter your choice: 3

Queue elements: 10 20

--- Queue Menu ---

Enter your choice: 2

10 deleted from queue.

--- Queue Menu ---

Enter your choice: 2

20 deleted from queue.

--- Queue Menu ---

Enter your choice: 2

Queue Underflow! Cannot delete.

--- Queue Menu ---

Enter your choice: 4

Exiting program.

Result :

Thus, the program to implement **Queue operations using an array** was successfully **designed, developed, and executed** in C.